

CONTACT INFORMATION	<i>E-mail:</i> vt286@cornell.edu <i>Website:</i> vstumuluri.github.io	
EDUCATION	Cornell University , Ithaca, NY August 2026 – Present Ph.D. Student, Applied Mathematics University of North Carolina , Chapel Hill, NC August 2022 – May 2026 B.S. Statistics & Analytics, Mathematics - Undergraduate thesis (advised by Sayan Banerjee): <i>Root-Finding and Opinion Dynamics in Random Growing Networks</i>; awarded highest honors in Statistics and Analytics - Received the Statistics and Analytics Award - Graduated with highest distinction and inducted into Phi Beta Kappa	
RESEARCH EXPERIENCE	University of North Carolina , Chapel Hill, NC August 2025 – Present Root-finding algorithms and opinion dynamics in random growing networks. Undergraduate honors thesis supported by the RTG in Networks at the Department of Statistics & Operations Research . Mentored by Sayan Banerjee . North Carolina State University , Raleigh, NC May 2025 – Present Numerical linear algebra methods for optimal sensor placement in Bayesian inverse problems. Participant in the DRUMS REU at the Department of Mathematics and the Department of Statistics . Mentored by Arvind K. Saibaba and Hugo Díaz . University of North Carolina , Chapel Hill, NC August 2024 – May 2025 Optimization methods for low-rank matrix reconstruction from Gaussian measurements. Supported by the RTG in Networks at the Department of Statistics & Operations Research . Mentored by Mike O'Neill . Montana State University , Bozeman, MT May 2024 – August 2024 Single-source shortest paths on parametrically weighted families of graphs. Participant in the Algorithms & Optimization REU at the Gianforte School of Computing . Mentored by Brittany Terese Fasy and Binhai Zhu .	
OTHER EXPERIENCE	Chessvia , Carrboro, NC April 2023 – August 2023 Software engineering intern. Worked on development and feature enhancement of Chessvia website. Developed and refined AI model that analyzes users' online chess accounts to create training plans and opening repertoires and match students with suitable coaches.	
PUBLICATIONS	[1] Gabriela Ramon, Geena Sarnoski, Vasishtha Tumuluri, Hugo Díaz, and Arvind K. Saibaba. Multifidelity sensor placement in Bayesian state estimation problems. <i>Applied Mathematics for Modern Challenges</i> , 2026, 9: 1-29. doi: 10.3934/ammc.2026006. Available at arXiv:2602.07269 . [2] Jacob Sriraman, Eli Barton, Brittany Terese Fasy, David L. Millman, Brendan Mumey, Nate Rengo, Braeden Sopp, Vasishtha Tumuluri, Binhai Zhu. Parametric Shortest Paths in a Linearly Interpolated Graph. <i>Accepted, Young Researchers Forum, Computational Geometry Week 2026</i> , available at arXiv:2604.08892 .	
TALKS	[3] Gabriela Ramon, Geena Sarnoski, Vasishtha Tumuluri , Hugo Díaz, and Arvind K. Saibaba. "Multifidelity Sensor Placement in Bayesian State Estimation Problems." Triangle Computational and Applied Mathematics Symposium, North Carolina State University, November 8-9, 2025, Raleigh, NC. <i>Lightning Talk</i> .	

	[4] Vasishta Tumuluri. "Gambler's Ruin: A First Step into Random Walks." MathGEMS at Carolina Math Club, September 16, 2025, Chapel Hill, NC. [5] Vasishta Tumuluri. "Fallacies, Flaws, and Howlers." MathGEMS at Carolina Math Club, March 4, 2025, Chapel Hill, NC.
POSTERS	[6] Gabriela Ramon, Geena Sarnoski, Vasishta Tumuluri, Hugo Díaz, and Arvind K. Saibaba. "Multifidelity Sensor Placement in Bayesian State Estimation Problems." Joint Mathematics Meetings, American Mathematical Society, January 4-7, 2026, Washington, D.C. [7] Gabriela Ramon, Geena Sarnoski, Vasishta Tumuluri, Hugo Díaz, and Arvind K. Saibaba. "Multifidelity Sensor Placement in Bayesian State Estimation Problems." Triangle Computational and Applied Mathematics Symposium, North Carolina State University, November 8-9, 2025, Raleigh, NC. [8] Brittany Fasy, Brendan Mumey, Nate Rengo, Braeden Sopp, Vasishta Tumuluri, and Binhai Zhu. "Parametric Shortest Paths on Affine Combinations of Weighted Directed Graphs." Summer Research Symposium, Montana State University, August 1, 2024, Bozeman, MT.
TEACHING	Cornell University, Ithaca, NY Teaching Assistant • MATH 4330 (Honors Linear Algebra) (Fall 2026) University of North Carolina, Chapel Hill, NC Undergraduate Teaching Assistant • MATH 130 (Precalculus Mathematics) (Spring 2026) • MATH 521 (Advanced Calculus I) (Fall 2024)
GRANTS	(i) American Mathematical Society undergraduate travel grant for research presentation at the Joint Mathematics Meetings 2026 (funded by NSF-DMS-2015553). Total award amount: \$1,000. <i>Awarded.</i>
TECHNOLOGY	Proficient: Python, Julia, $\text{\LaTeX}$ Familiar: R, MATLAB, Java